Blockchain Experiment 8

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Aim: To configure a cryptocurrency wallet using MetaMask and integrate it with a Decentralized Application (DApp) designed for optimal and secure Banking Transactions on the blockchain.

Theory:

1. MetaMask as a Web3 Provider

MetaMask acts as a bridge between web browsers and Ethereum-based blockchain networks. It injects a global Ethereum API object (window.ethereum) into the browser context, allowing decentralized web applications to request user wallet authorization, retrieve public addresses, and prompt users to sign banking transactions using their private keys.

2. Banking DApp Architecture & Front-End Integration

To handle banking workflows (depositing, withdrawing, and transferring Ether safely) from a user interface, client-side applications utilize libraries like Ethers.js or Web3.js:

- *Provider*: An abstraction for reading blockchain state data (e.g., querying account balances or checking contract treasury funds without gas fees).
- *Signer*: Represents an authorized user account capable of signing value-bearing transactions (e.g., depositing Ether or transferring funds).
- *Contract Instance*: A JavaScript representation of the smart contract initialized via its Deployed Address and ABI (Application Binary Interface).

3. Reentrancy Protection & Gas Optimization in Banking Smart Contracts

To ensure banking operations are executed in a safe and optimal manner:

- *Checks-Effects-Interactions Patter*n: State variables are updated in storage before external Ether transfers occur to prevent reentrancy attacks.
- *Gas-Efficient State Updates*: Reusable function execution logic avoids unnecessary computations and transaction failures on the EVM.

Program Code:

OptimalBankingDApp.sol

```
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.0;

/**
 * @title OptimalBankingDApp
 * @dev Secure and gas-optimized decentralized banking contract.
 */
contract OptimalBankingDApp {
    mapping(address => uint256) private balances;

    event Deposit(address indexed account, uint256 amount);
    event Withdrawal(address indexed account, uint256 amount);
    event Transfer(address indexed sender, address indexed receiver, uint256 amount);

    // Deposit Ether into the banking contract
    function deposit() external payable {
        require(msg.value > 0, "Deposit amount must be greater than zero");
        balances[msg.sender] += msg.value;
        emit Deposit(msg.sender, msg.value);
    }

    // Withdraw Ether safely using Checks-Effects-Interactions pattern
    function withdraw(uint256 _amount) external {
        require(_amount > 0, "Withdrawal amount must be greater than zero");
        require(balances[msg.sender] >= _amount, "Insufficient bank balance");

        // Effect
        balances[msg.sender] -= _amount;

        // Interaction
        (bool success, ) = payable(msg.sender).call{value: _amount}("");
        require(success, "Ether transfer failed");

        emit Withdrawal(msg.sender, _amount);
    }

    // Optimal peer-to-peer internal transfer between bank accounts
    function transferFunds(address payable _to, uint256 _amount) external {
        require(_to != address(0), "Invalid recipient address");
        require(_to != msg.sender, "Cannot transfer to self");
        require(_amount > 0, "Transfer amount must be greater than zero");
        require(balances[msg.sender] >= _amount, "Insufficient balance");

        balances[msg.sender] -= _amount;
        balances[_to] += _amount;

        emit Transfer(msg.sender, _to, _amount);
    }

    // Read user balance
    function getBalance(address _account) external view returns (uint256) {
        return balances[_account];
    }
}
```

Index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport"
content="width=device-width,
initial-scale=1.0">
    <title>Decentralized Banking
DApp</title>
    <style>
        body { font-family: 'Segoe
UI', Tahoma, Geneva, Verdana,
sans-serif; margin: 30px;
background-color: #f0f2f5; }
        .card { background: white;
padding: 25px; border-radius: 10px;
max-width: 500px; box-shadow: 0 4px
12px rgba(0,0,0,0.1); margin: auto;
}
        button { background-color:
#107c41; color: white; border: none;
padding: 10px 16px; font-size: 14px;
border-radius: 6px; cursor: pointer;
font-weight: 600; margin-top: 5px; }
        button:hover {
background-color: #0b5a2f; }
        .btn-connect {
background-color: #f6851b; }
        .btn-connect:hover {
background-color: #e2761b; }
        input { padding: 8px 12px;
width: calc(100% - 28px); margin:
6px 0 12px 0; border: 1px solid
#ccc; border-radius: 6px; }
        .status { margin-top: 10px;
font-size: 14px; color: #333;
word-break: break-all; }
        .balance-box { background:
#e8f5e9; padding: 12px;
border-radius: 6px; font-weight:
bold; font-size: 16px; color:
#1b5e20; margin: 15px 0; }
    </style>
    <script
src="https://cdnjs.cloudflare.com/aj
ax/libs/ethers/5.7.2/ethers.umd.min.
js"></script>
</head>
<body>

<div class="card">
    <h2>Decentralized Bank
Portal</h2>

    <button class="btn-connect"
id="connectBtn"
onclick="connectWallet()">Connect
MetaMask Wallet</button>
    <div class="status"
id="accountStatus">Wallet Status:
Not Connected</div>

    <div class="balance-box"
id="bankBalanceDisplay">Vault
Balance: 0.0 ETH</div>

    <hr style="margin: 20px 0;">

    <h3>Banking Operations</h3>

    <label><b>Deposit
ETH:</b></label>
    <input type="number" step="0.01"
id="depositInput"
placeholder="Amount in ETH">
    <button
onclick="depositETH()">Deposit to
Bank</button>

    <br><br>
    <label><b>Withdraw
ETH:</b></label>
    <input type="number" step="0.01"
id="withdrawInput"
placeholder="Amount in ETH">
    <button
onclick="withdrawETH()">Withdraw to
Wallet</button>

    <br><br>
    <label><b>Transfer Funds
(Internal):</b></label>
    <input type="text"
id="recipientInput"
placeholder="Recipient Address
(0x...)">
```

```
    <input type="number" step="0.01"
id="transferInput"
placeholder="Amount in ETH">
    <button
onclick="transferETH()">Transfer
Funds</button>

    <div class="status"
id="txStatus" style="margin-top:
15px; font-weight: 600;"></div>
</div>

<script>
    const contractAddress =
"0xDA0bab807633f07f013f94DD0E6A4F96F
8742B53";

    const contractABI = [
        "function deposit() external
payable",
        "function withdraw(uint256
_amount) external",
        "function
transferFunds(address payable _to,
uint256 _amount) external",
        "function getBalance(address
_account) external view returns
(uint256)"
    ];

    let provider, signer, contract,
userAccount;

    async function connectWallet() {
        if (typeof window.ethereum
!== 'undefined') {
            try {
                const accounts =
await window.ethereum.request({
method: 'eth_requestAccounts' });
                userAccount =
accounts[0];

document.getElementById('accountStat
us').innerText = "Account: " +
userAccount;

                provider = new
ethers.providers.Web3Provider(window
.ethereum);
                signer =
provider.getSigner();
                contract = new
ethers.Contract(contractAddress,
contractABI, signer);

                updateBalance();
            } catch (err) {

console.error("Connection Error:",
err);
            }
        } else {
            alert("MetaMask
extension not detected!");
        }
    }

    async function updateBalance() {
        if (!contract ||
!userAccount) return;
        try {
            const rawBalance = await
contract.getBalance(userAccount);
            const formatted =
ethers.utils.formatEther(rawBalance)
;

document.getElementById('bankBalance
Display').innerText = `Vault
Balance: ${formatted} ETH`;
        } catch (err) {
            console.error("Balance
Fetch Error:", err);
        }
    }

    async function depositETH() {
        const val =
document.getElementById('depositInpu
t').value;
        if (!val || val <= 0) return
alert("Enter a valid ETH amount");

        try {
```

```
document.getElementById('txStatus').
innerText = "Processing Deposit...";
            const tx = await
contract.deposit({ value:
ethers.utils.parseEther(val) });
            await tx.wait();

document.getElementById('txStatus').
innerText = "Deposit Successful! TX
Hash: " + tx.hash;
            updateBalance();
        } catch (err) {
            console.error(err);

document.getElementById('txStatus').
innerText = "Deposit failed or
cancelled.";
        }
    }

    async function withdrawETH() {
        const val =
document.getElementById('withdrawInp
ut').value;
        if (!val || val <= 0) return
alert("Enter a valid ETH amount");

        try {

document.getElementById('txStatus').
innerText = "Processing
Withdrawal...";
            const tx = await
contract.withdraw(ethers.utils.parse
Ether(val));
            await tx.wait();

document.getElementById('txStatus').
innerText = "Withdrawal Successful!
TX Hash: " + tx.hash;
            updateBalance();
        } catch (err) {
            console.error(err);

document.getElementById('txStatus').
innerText = "Withdrawal failed.";
        }
    }

    async function transferETH() {
        const recipient =
document.getElementById('recipientIn
put').value;
        const val =
document.getElementById('transferInp
ut').value;
        if (!recipient || !val ||
val <= 0) return alert("Enter valid
recipient and amount");

        try {

document.getElementById('txStatus').
innerText = "Processing Internal
Transfer...";
            const tx = await
contract.transferFunds(recipient,
ethers.utils.parseEther(val));
            await tx.wait();

document.getElementById('txStatus').
innerText = "Transfer Successful! TX
Hash: " + tx.hash;
            updateBalance();
        } catch (err) {
            console.error(err);

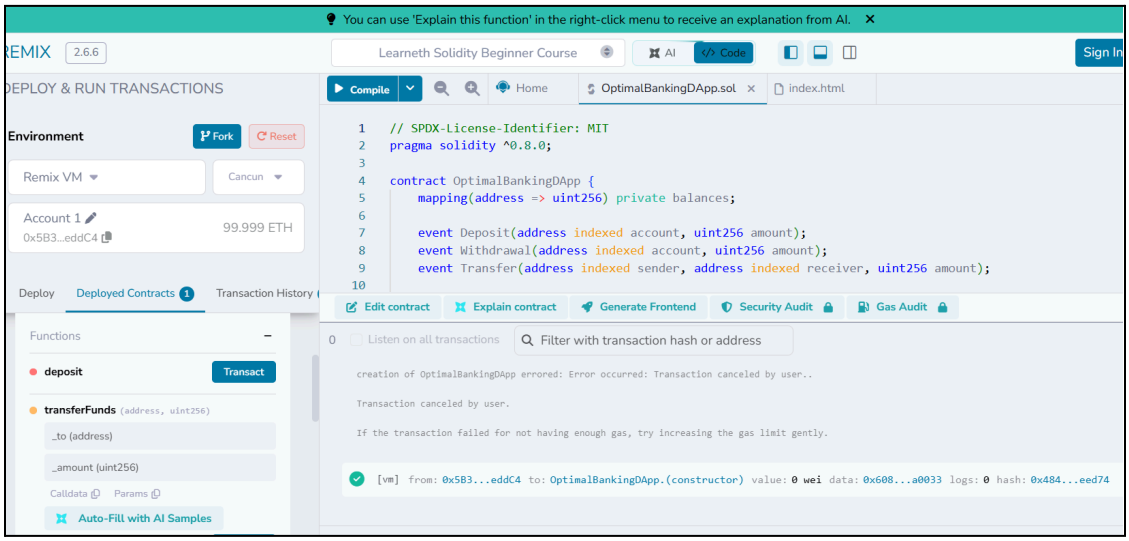
document.getElementById('txStatus').
innerText = "Transfer failed.";
        }
    }
</script>

</body>
</html>
```

Implementation:

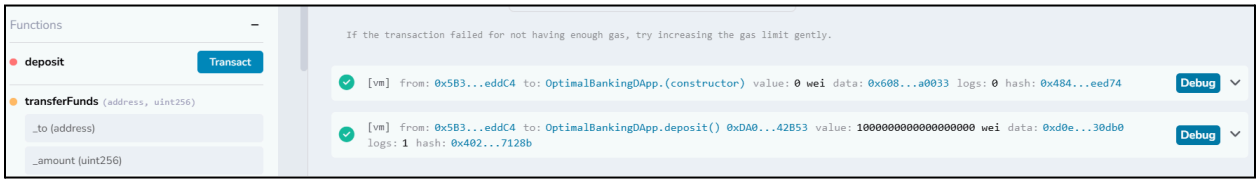
Step 1: Smart Contract Compilation & Deployment in Remix

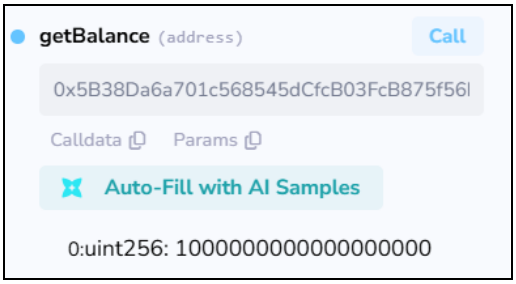
1. Created OptimalBankingDApp.sol in Remix IDE under Experiment_8_Blockchain.
2. Compiled the contract using Solidity compiler 0.8.0.
3. Selected Remix VM (Cancun) as the environment and deployed the contract.
4. Copied the deployed contract address: 0xDA0bab807633f07f013f94DD0E6A4F96F8742B53.



Step 2: Testing Contract Functions in Remix

1. Deposit Function: Set Value to 1000000000000000000 wei (1 ETH) and executed deposit(). Verified block mining in console.
2. State Balance Check: Called getBalance(0x5B3...eddC4) which returned 1000000000000000000 wei.
3. Internal Transfer: Called transferFunds with recipient 0xAb8483F64d9C6d1EcF9b849Ae677dD3315835cb2 and amount 500000000000000000 wei (0.5 ETH)

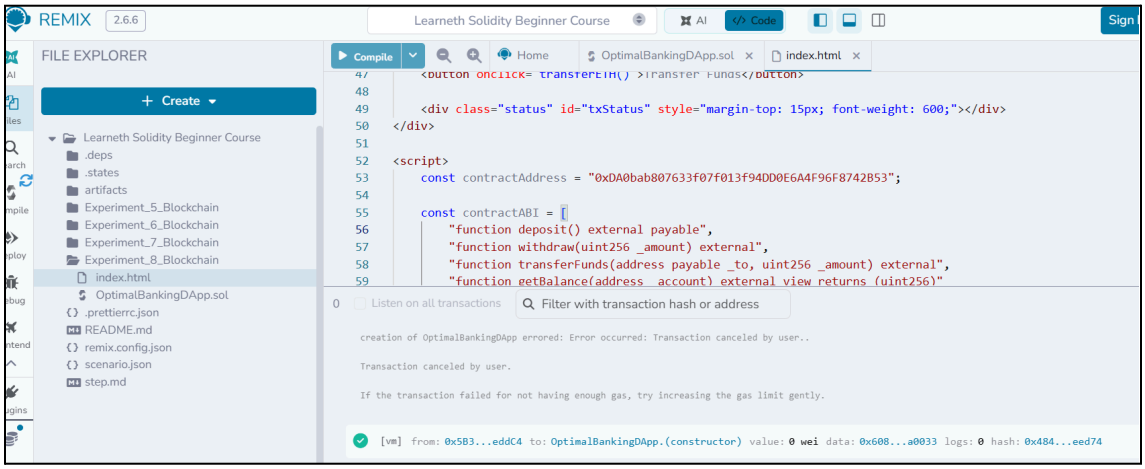


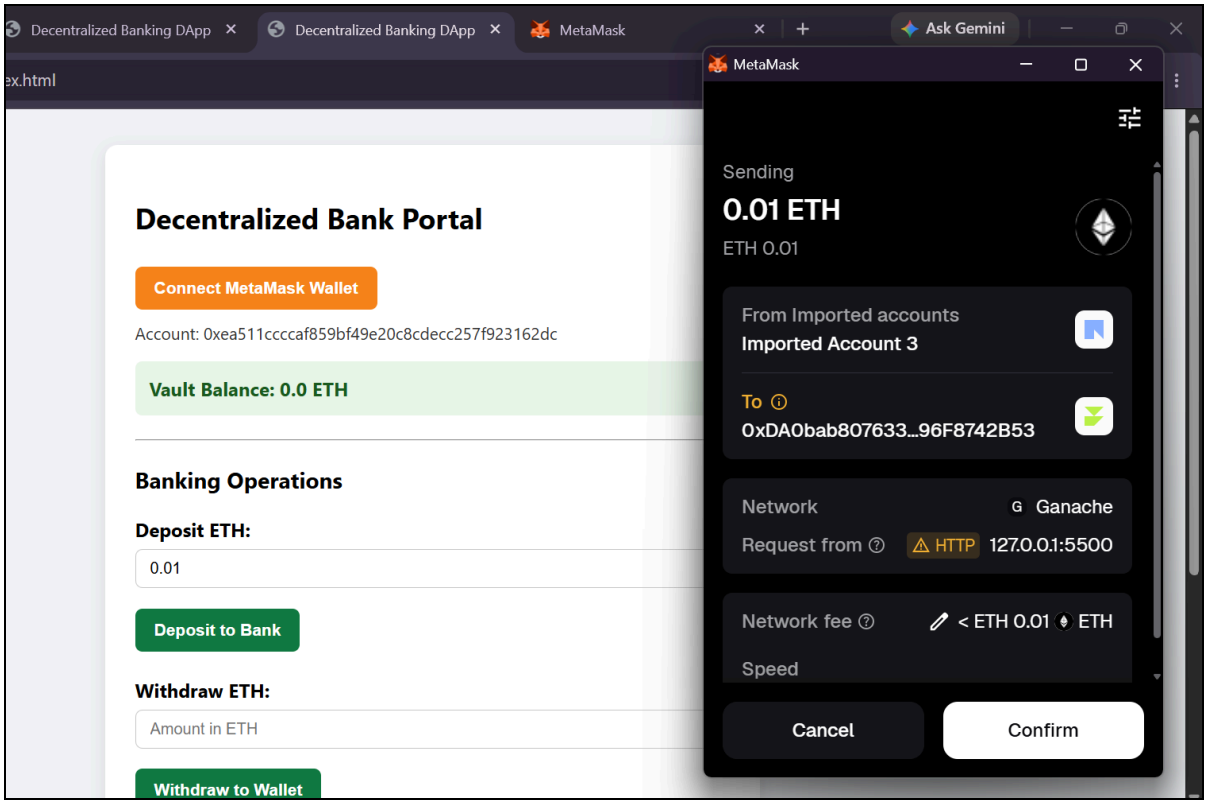




Step 3: Web3 Integration with MetaMask & Front-End UI

1. Configured index.html with the deployed contract address 0xDA0bab807633f07f013f94DD0E6A4F96F8742B53 and ABI methods.
2. Launched the DApp locally via VS Code Live Server at http://127.0.0.1:5500/index.html[cite: 12, 20].
3. Clicked Connect MetaMask Wallet, connecting imported account 0xea511ccccaf859bf49e20c8cdecc257f923162dc.
4. Entered 0.01 ETH in the deposit section and triggered Deposit to Bank.
5. Approved the transaction prompt in MetaMask connected to the local Ganache network.





Conclusion:

A decentralized Banking Application (DApp) was successfully implemented, deployed, and integrated with MetaMask wallet. By using Ethers.js and injected Web3 provider instances (window.ethereum), seamless interaction was established between client-side JavaScript and smart contracts on the Ethereum network. Practical operations including wallet account linking, state reading without gas fees, and signing transactions for contract balance deposits were verified.